

Threat Composition



Fire 31%

Deforestation 45%

Legal 13%

Other 11%

Permit Pipeline

ACTIVE: 34



Avg. Processing Time

4.2 days

Approval Rate (30d)

68.4%

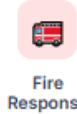
Pending Review

8 permits

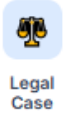
High-Priority

3 urgent

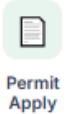
Quick Actions



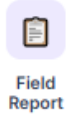
Fire
Response



Legal
Case



Permit
Apply



Field
Report

SYSTEM HEALTH

NASA FIRMS Feed

98%

OK

GFW GLAD Feed

95%

OK

AI Fuser Engine

99%

OK

OWM Climate API

88%

Lag

7-Day Climate & Fire Risk Forecast

OWM - HAZARA DIVISION

MON



28°C

HIGH

TUE



31°C

HIGH

WED



33°C

CRIT

THU



24°C

MED

FRI



22°C

LOW

SAT



26°C

MED

SUN



30°C

HIGH

Citizen Forest Awareness Hub

Public Intelligence dashboard — Hazara Division: Abbottabad, Mansehra, Kohistan, Batagram, Haripur

Current Forest Status

 Active Fire Alerts

75

↑ NASA FIRMS · live

 Deforestation Alerts

124

↑ GFW GLAD · 30d

 Fused Threat Zones

24

↑ AI Correlation

 Satellite Visibility

85%

↑ Weather + Sensor

District Threat Levels

 Abbottabad

CRITICAL · 24 events

 Mansehra

CRITICAL · 28 events

 Kohistan

HIGH · 6 events

 Batagram

CRITICAL · 8 events

 Haripur

CRITICAL · 8 events

Live Alert Feed

 FIRE — 75 thermal anomalies active (NASA FIRMS · Hazara)

 DEFORESTATION — 124 canopy loss events confirmed (GFW)

▼  Satellite Intelligence Health

Visibility

85%

Propensity Zones

54

↑ 30-day forecast

NASA FIRMS

STABLE

GFW Status

STABLE

Activate Windows
Go to Settings to activate Windows.



Environmental Monitoring

Live forest surveillance with tactical GIS layering, fire detection, and deforestation intelligence.

▼ Military-Grade Data Integrity Panel (LIVE)

🔴 NASA FIRMS (Fire)

Status: **STABLE**

Sync: 13:17:39

Raw Hits: 75

Clusters: 14

Strict Hazara Operational Bounding Box Enforced

🟢 GFW (Deforestation)

Status: **STABLE**

Sync: 13:17:39

Raw Hits: 124

Clusters: 16

🔴 AI Fuser Engine

Status: **ONLINE**

Fusion Dist: 10 km

Deduplication Active

Sync



High-Integrity Tactical Intelligence Panel

🔴 Raw Fire Hits

75

↑ 14 Clusters

👁 Systemic Visibility

85.0%

↑ Derived (Weather + Sensor)

🟢 Raw Tree Hits

124

↑ 16 Clusters

🟡 Propensity Zones

61

↑ 30-Day Forecast Active

🔴 Fused Threats

24

↑ 12.1% Density

🔴 Threat Density

12.1%

↑ Fused / Raw Signals

🟡 30-Day Threat Vectors

61 Active

↑ Forecasted Vulnerability

👤 Total Events

124

↑ In Fusion Window

▼ Intelligence Logic & Visibility Breakdown

Current intelligence accuracy is calculated against an internal Ground-Truth baseline

Activate Windows
Go to Settings to activate Windows.



Report a Forest Issue

Your report goes directly to DFO via email and Slack. Anonymous submissions accepted.

Report type



Fire / Smoke



Location (District/Tehsil/Village)

e.g., Abbottabad, Sherwan

Description

Describe what you observed — time, vehicles, extent of damage...

Your contact (optional)

Phone or email for follow-up



Submit Report to DFO



How to Protect Hazara's Forests



Spot a fire? Call DFO immediately. Note location and wind direction. Do not approach.



Illegal logging signs Fresh stumps, chainsaw sounds at night, unmarked timber trucks.



Plant native trees Deodar, Blue Pine, Kail — contact DFO for free saplings.

KPK Forest Ordinance 2002: Unauthorized felling or timber transport without a permit is punishable by fine up to Rs. 50,000 and/or 2 years imprisonment.

DFO Emergency Line: +92-349-599-4503 | Email: alihinaali2022@gmail.com

Activate Windows

Go to Settings to activate Windows.

1. Data ingestion layer

NASA FIRMS (MODIS/VIIRS) · Global Forest Watch · OpenWeatherMap

raw feeds

2. Processing & fusion layer

Fusion engine · zero-latency cache manager · 30-day predictor

fused, cached signals

3. Intelligence layer

Agentic orchestrator · FAISS vector DB · Neo4j graph DB · hallucination guard

5 agents: law, climate, monitoring, judiciary, permit

grounded answers

4. Presentation layer (Streamlit)

portal.py command & control portal, plus 8 modules

Dashboard/Chat/Legal/Judiciary/Monitoring/Predict/Ops/Citizen/Permit

operator actions

5. Communication & enforcement layer

Slack alerts · email relay · PDF evidence engine · blockchain ledger



Environmental Monitoring

Live forest surveillance with tactical GIS layering, fire detection, and deforestation intelligence.

▼ Military-Grade Data Integrity Panel (LIVE)

● NASA FIRMS (Fire)

Status: **STABLE**

Sync: 13:17:39

Raw Hits: 75

Clusters: 14

Strict Hazara Operational Bounding Box Enforced

● GFW (Deforestation)

Status: **STABLE**

Sync: 13:17:39

Raw Hits: 124

Clusters: 16

🌀 AI Fuser Engine

Status: **ONLINE**

Fusion Dist: 10 km

Deduplication Active

Sync



High-Integrity Tactical Intelligence Panel

🔥 Raw Fire Hits

75

↑ 14 Clusters

👁 Systemic Visibility

85.0%

↑ Derived (Weather + Sensor)

● Raw Tree Hits

124

↑ 16 Clusters

👤 Propensity Zones

61

↑ 30-Day Forecast Active

🌀 Fused Threats

24

↑ 12.1% Density

🔥 Threat Density

12.1%

↑ Fused / Raw Signals

👤 30-Day Threat Vectors

61 Active

↑ Forecasted Vulnerability

👤 Total Events

124

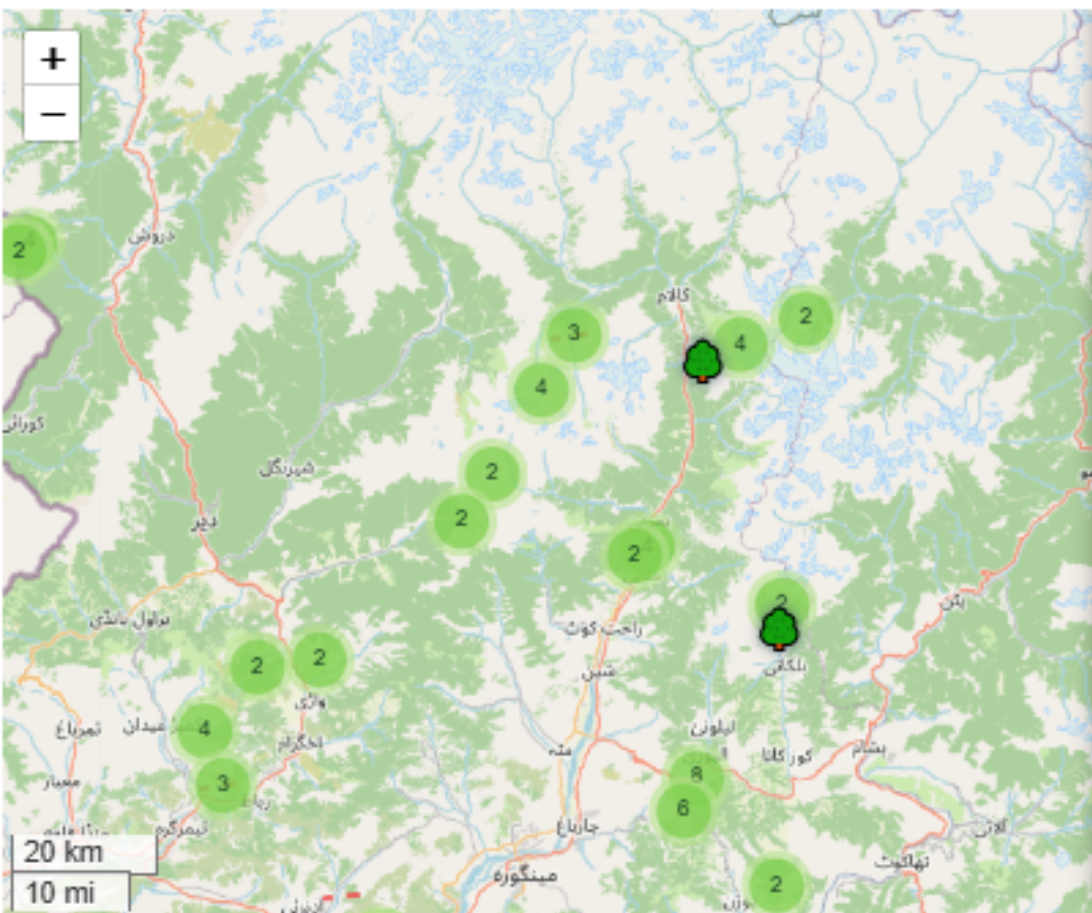
↑ In Fusion Window

▼ 🧩 Intelligence Logic & Visibility Breakdown

Current intelligence accuracy is calculated against an internal Ground-Truth baseline

Activate Windows
Go to Settings to activate Windows.

Tactical Geo-Intelligence Map



DEFORESTATION DETECTED

MEDIUM

ID: **GL-EB2CD1**

SOURCE: FUSED-INTEL

DETECTED: 2026-06-16 00:00 UTC

LOCAL AREA: **Balakot (32)**

"Localized canopy loss signal identified in Balakot (32.0km). Verifying biomass displacement."

CLUSTER SIZE: 1 Signals

SENSOR DIVERSITY: 1 Sources

CARBON SEQUESTRATION AT RISK 331.5 MTCO2

Biomass: 275.25 | Annual Lost: 56.25

Est. Area: 5.0 ha

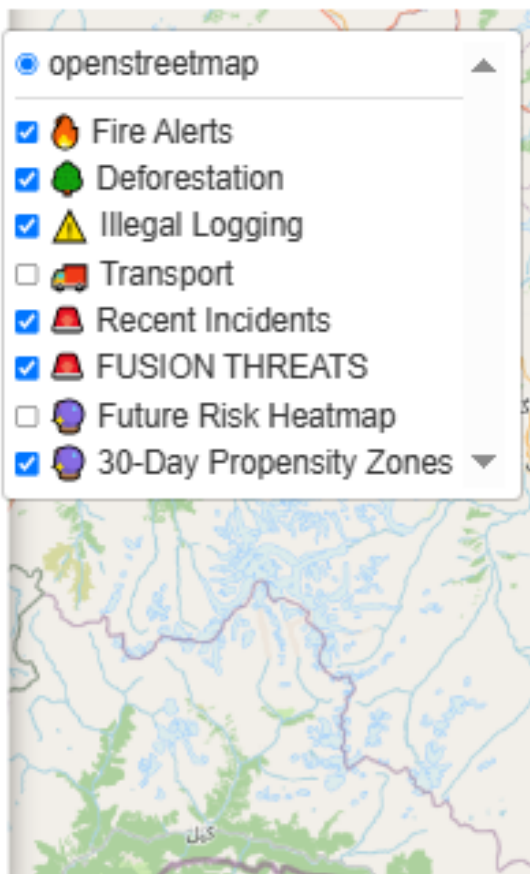
CONFIDENCE

24%

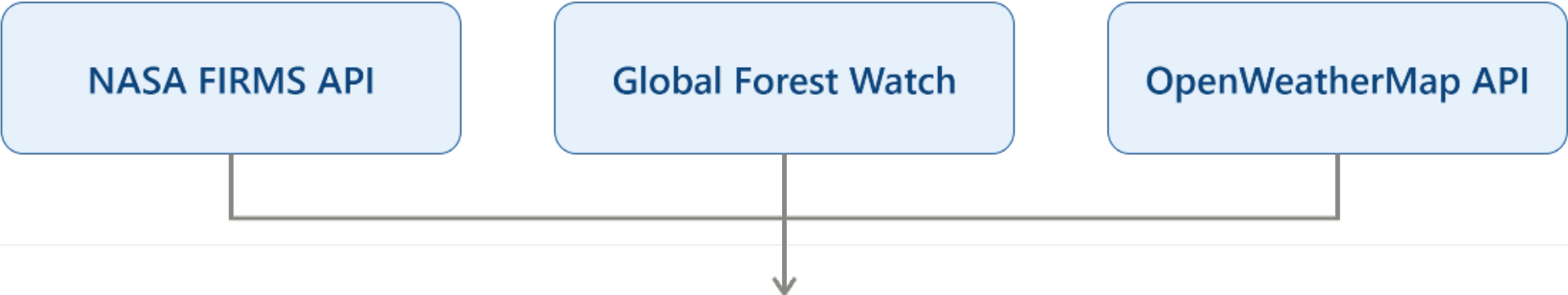
SYSTEM RECALL

100% (Verifiable
Performance)

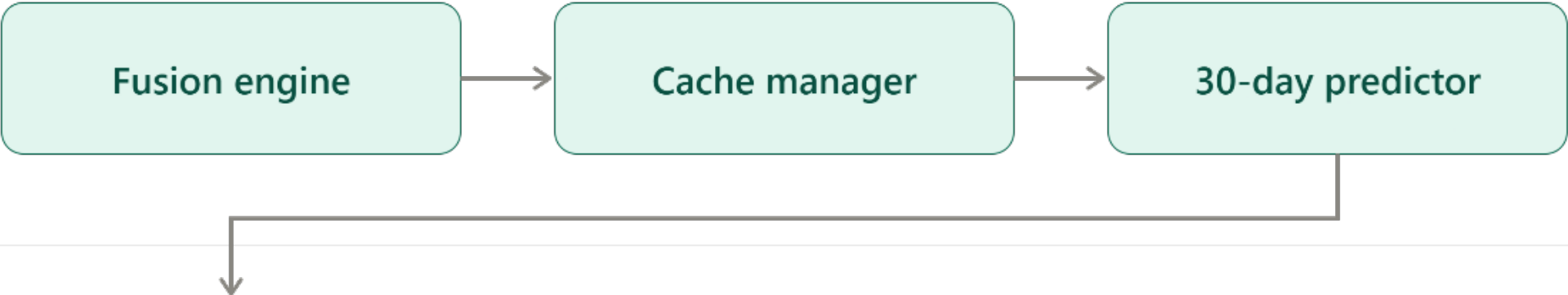
LINEAGE: FUSED-INTEL



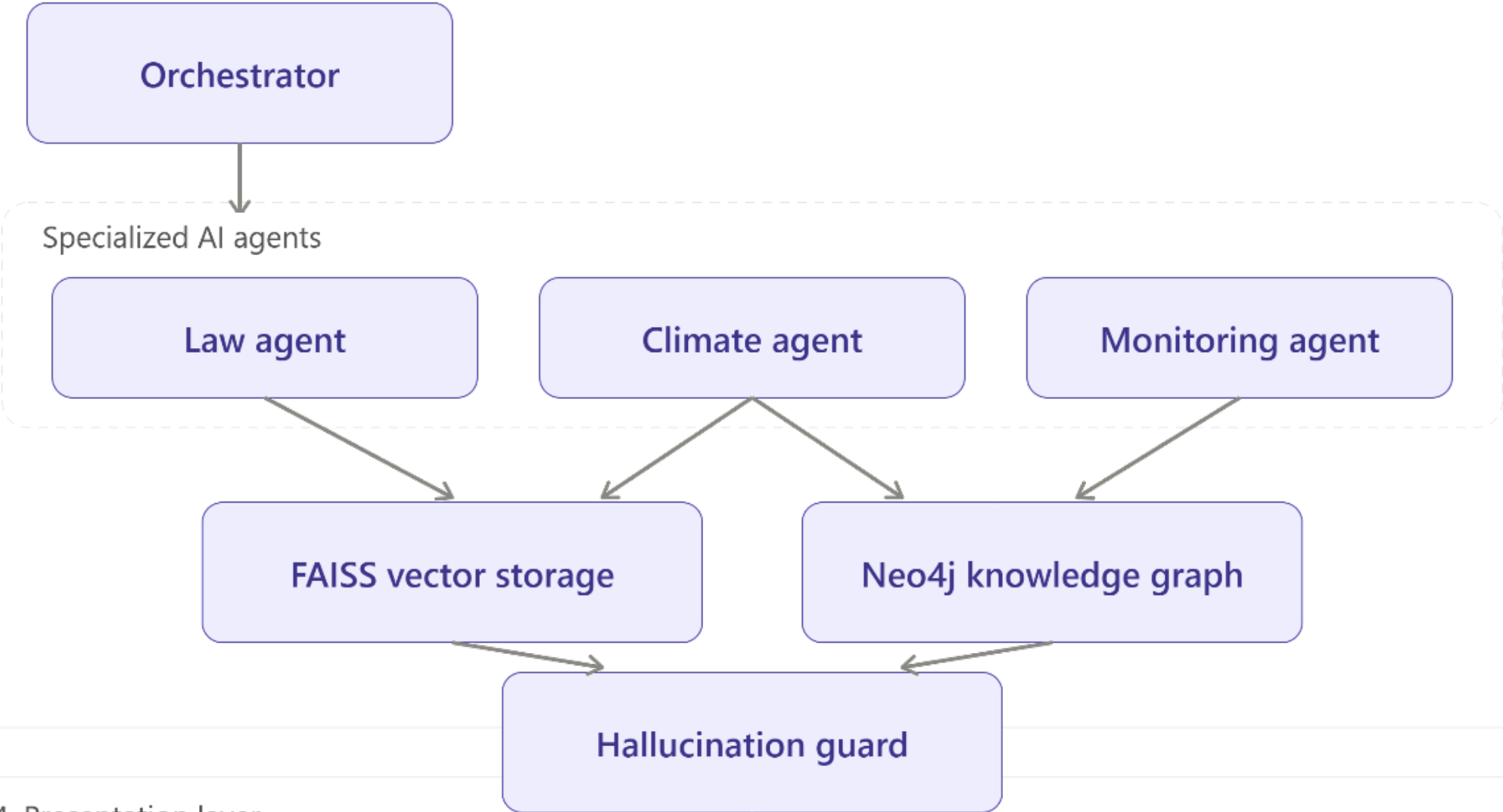
1. Data ingestion layer



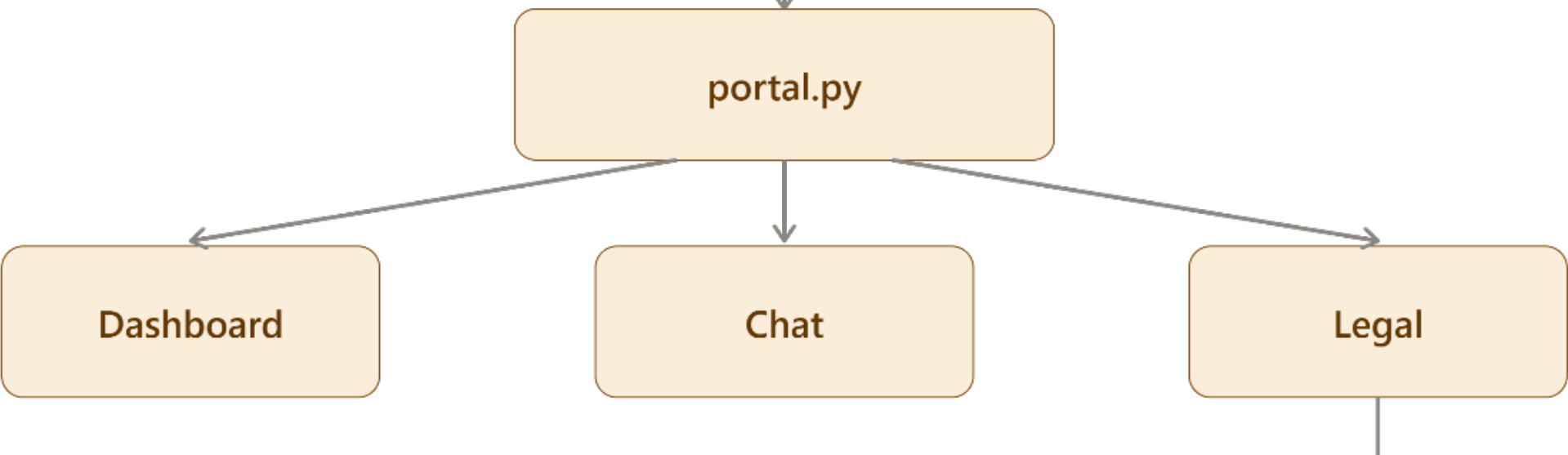
2. Processing & fusion layer



3. Intelligence layer

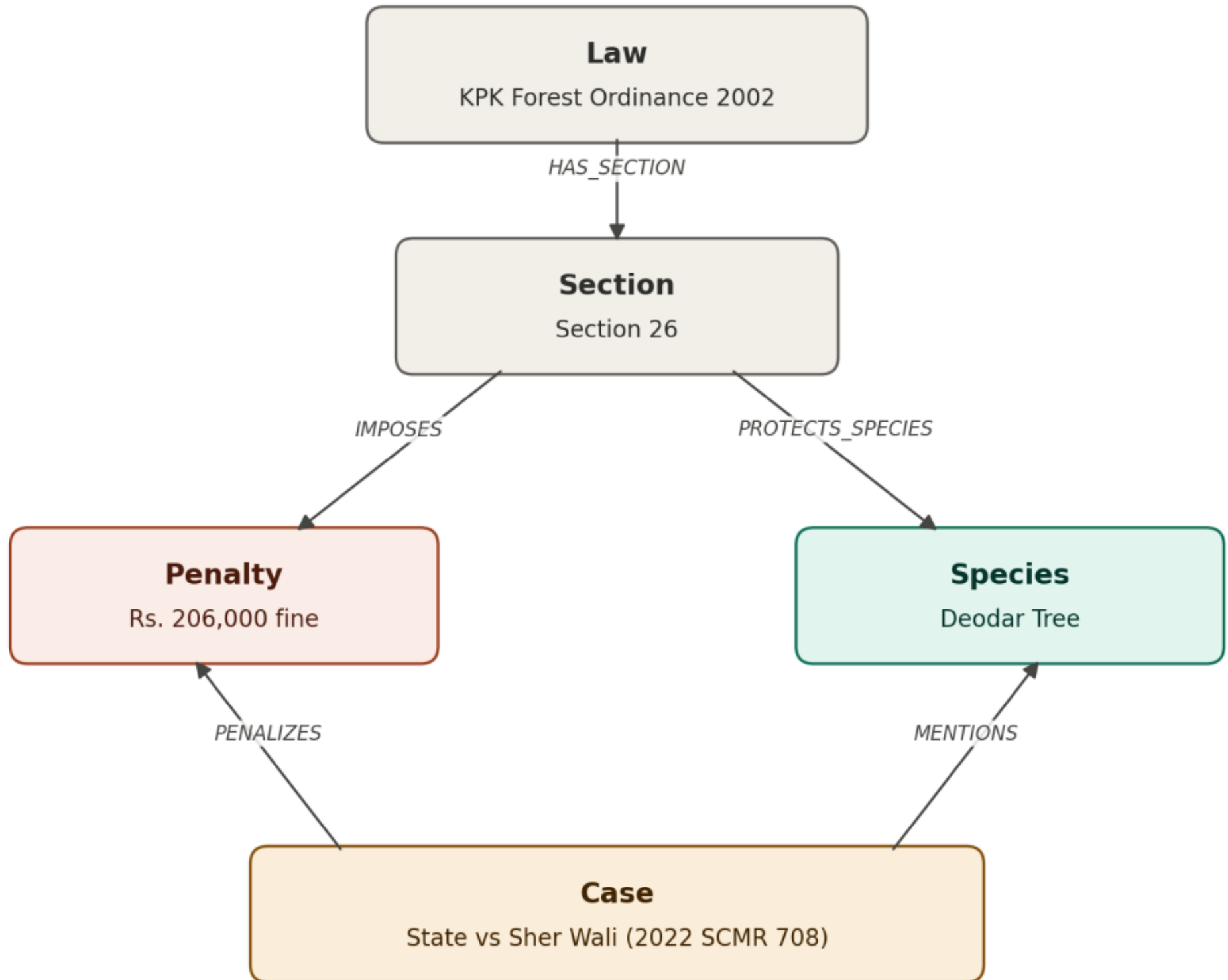


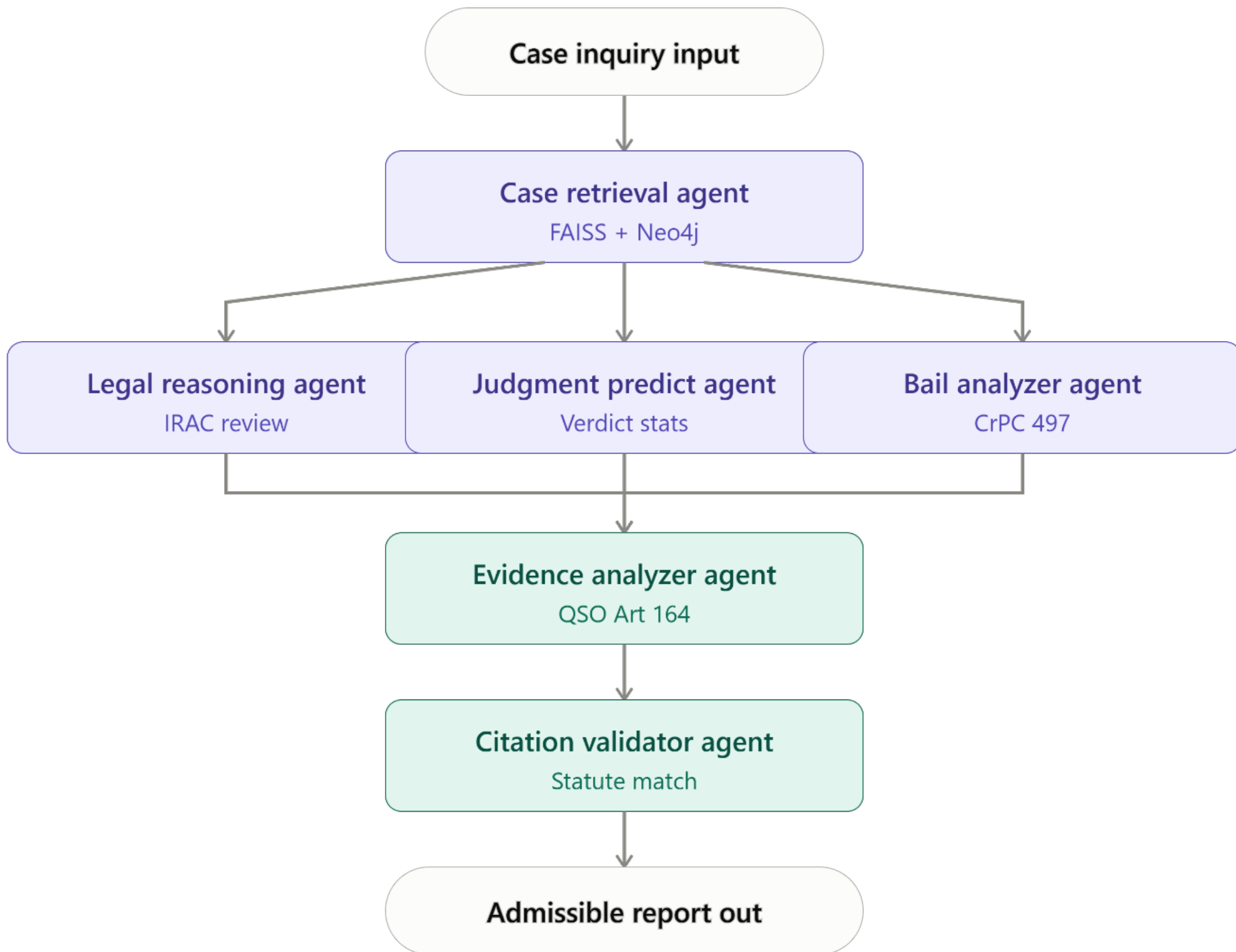
4. Presentation layer



5. Communication layer









GreenBot AI Assistant

Forest Intelligence · Legal · Monitoring

● Online

What is the current fire threat level in
Mansehra district?



DFO Hazara · 2 min ago



🔥 CRITICAL ALERT — Mansehra District

Based on real-time NASA FIRMS data and AI analysis:

- **Active fire hits:** 9 confirmed (VIIRS 375m)
- **Threat level:** **CRITICAL**
- **AI Confidence:** 92%
- **Most affected:** Block-7 (Darband area)
- **Wind direction:** SW → NE at 24 km/h

🌡️ Temperature: 33°C | 💧 Humidity: 18% | ⚡ FWI Index: 8.4/10

Recommended action: Immediate deployment of Patrol Unit P-01 and P-02. Aerial surveillance requested. DFO authorization required under §37 KP Forest Ordinance 2002.

GreenBot · 1 min ago

 Suggested Queries

 Today's fire alerts

 Logging penalties

 Active patrol units

 Deforestation hotspots

 Generate field report

 CO₂ risk assessment

Check permit status

 Active Context

Knowledge Bases Loaded:

 KP Forest Ordinance 2002

 Forest Act 1927

 NASA FIRMS Live

 GFW GLAD

 OWM Weather

 Hazara GIS Data

Show me the recent satellite evidence for forest crimes in Abbottabad



DFO Hazara · Just now



Analysis Report

Query: Show me the recent satellite evidence for forest crimes in Abbottabad



GreenLawAI Autonomous Report

SATELLITE INTELLIGENCE: REAL-TIME DETECTION for Hazara

Today · 08:49 PKT



HIGH-PRIORITY ALERT DETECTED — Abbottabad District (Last 24 hrs)



Tactical Historical Intelligence



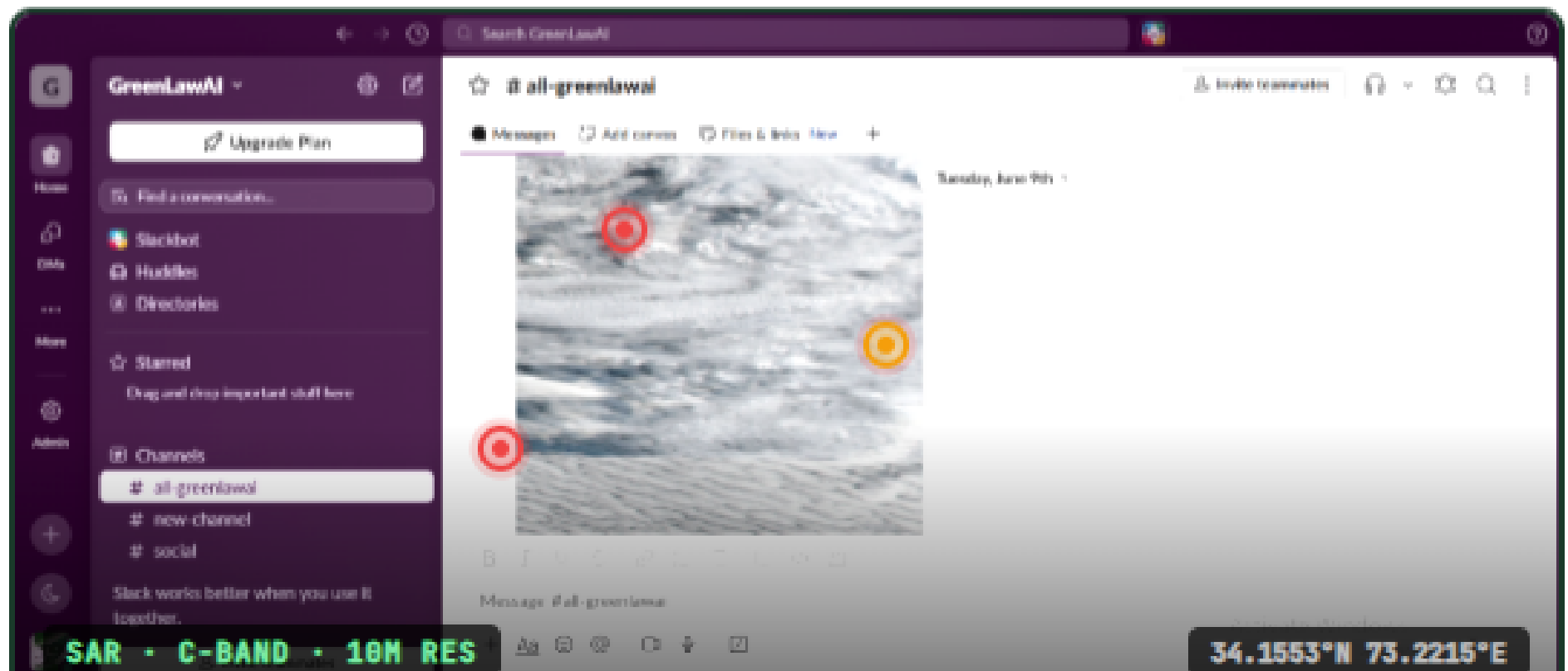
SATELLITE INTELLIGENCE: REAL-TIME DETECTION for Hazara

A high-priority alert has been detected within the last 24 hours via tactical sensors. The following data is drawn from **GFW GLAD Landsat-8 (30m)**, **Sentinel-1 SAR C-band**, and **NASA FIRMS VIIRS** cross-validation.

CURRENT ALERT DETECTED (Today):

- **Date:** 2026-06-09
- **Location:** [Abbottabad](#) · [Sector N-3](#)
- **Description:** Deforestation alert detected via GFW satellite. Confidence: **HIGH**
- **Alert Type:** **Illegal Timber Extraction** | **Area:** 0.82 ha
- **Source:** Global Forest Watch  [View Map](#)

Sentinel-1 SAR Imagery · Abbottabad Forest Reserve



GreenLawAI

Upgrade Plan

Find a conversation...

Slackbot

Huddles

Directories

Starred

Drag and drop important stuff here

Channels

- # all-greenlawal
- # new-channel
- # social

Slack works better when you use it together.

all-greenlawal

Messages

Add comment

Files & links

Search

Invite team member

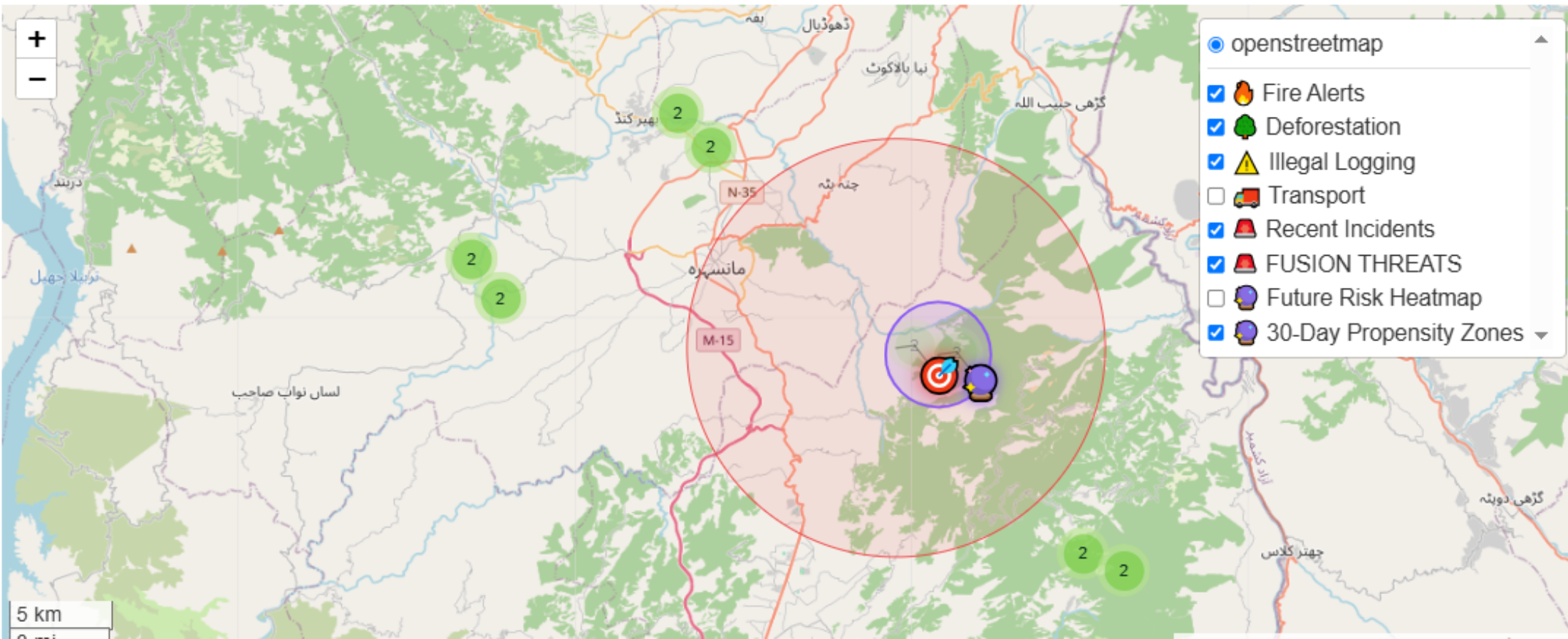
Sunday, June 9th

SAR - C-BAND - 10M RES

34.1553°N 73.2215°E

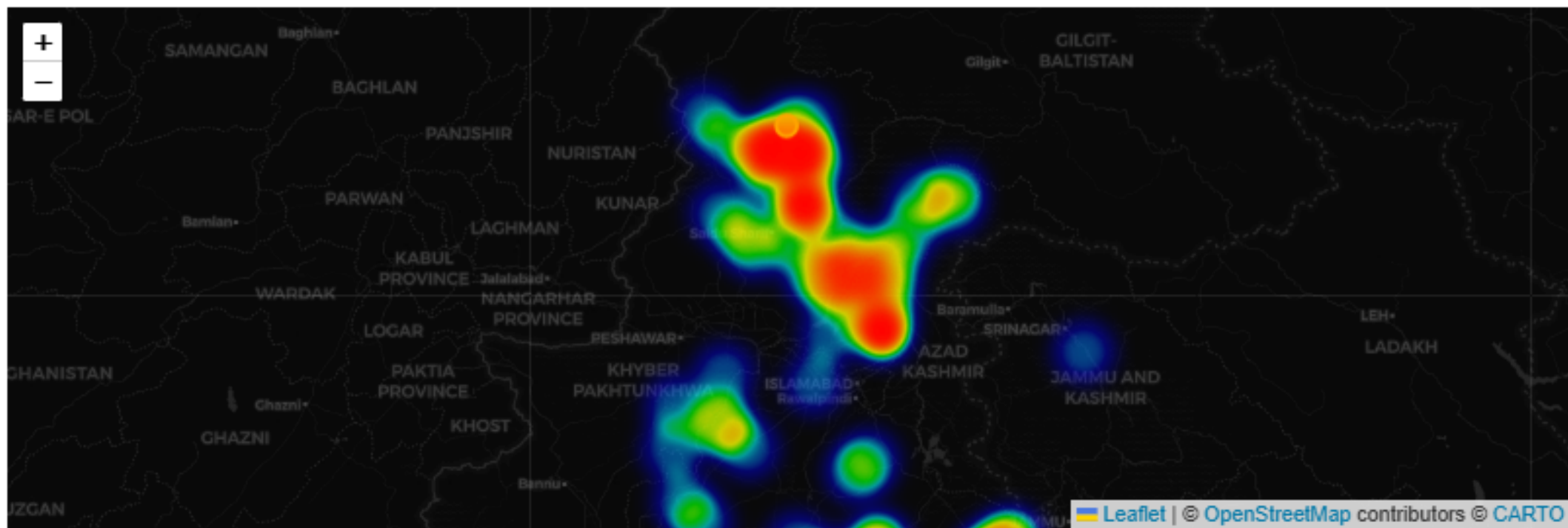


Tactical Geo-Intelligence Map

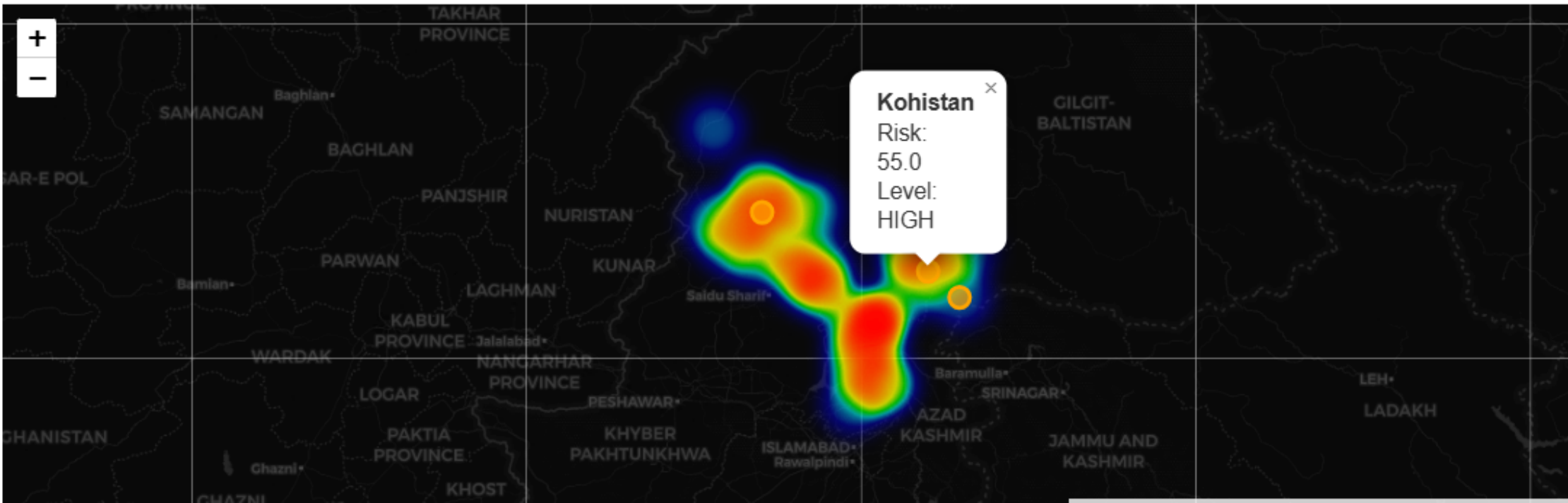




Deforestation Risk Heatmap



Deforestation Risk Heatmap





5-Year Carbon Projection

Best Case (-50% Policy Impact)

1,562,500 trees / 414,375t CO₂

↑ Managed

Current Trend (Unchanged)

3,125,000 trees / 828,750t CO₂

↑ Concerning

Worst Case (+30% Climate Driver)

4,062,500 trees / 1,077,375t CO₂

↑ Critical



30-Day Risk Propensity Forecast

Top 3 high-probability zones for illegal activity based on historical trends.

Risk Score

85-95%

↑ High Confidence



LOCATION: Abbottabad City (4.8km) (34.20, 73.20)



Rationale: High propensity in Abbottabad City (4.0km): 13 historical incidents. Climate surge factor: 0.49.

Confidence: STOCHASTIC (Trend Analysis) | Window: 30 Days

Risk Score

85-95%

↑ High Confidence



LOCATION: Mansehra City (3.4km) (34.30, 73.20)



Rationale: High propensity in Mansehra City (3.4km): 13 historical incidents. Climate surge factor: 0.49.

Confidence: STOCHASTIC (Trend Analysis) | Window: 30 Days

Risk Score

85-95%

↑ High Confidence



LOCATION: Abbottabad City (18.1km) (34.10, 73.40)



Rationale: High propensity in Abbottabad City (18.1km): 11 historical incidents. Climate surge factor: 0.49.

Confidence: STOCHASTIC (Trend Analysis) | Window: 30 Days

Activate Windows
Go to Settings to activate Windows.

INTELLIGENCE

 Dashboard 3 AI Chat

LEGAL

 Legal Analysis Judiciary 47

OPERATIONS

 Monitoring 23 Predict Operations

PUBLIC

 Citizen Reports 8

Permits

AI LEGAL RESEARCH ENGINE

Ask a Legal Question

Powered by KP Forest Ordinance 2002 · Forest Act 1927 · PEPA 1997 · Wildlife Act 1975

e.g., What is the penalty for illegal timber extraction under KP Forest Ordinance 2002?

Penalty for illegal logging

Forest encroachment offences

Wildlife protection Act 1975

Forest officer powers of arrest

Carbon credit legal framework

 Analyze Legal Analysis: Illegal Timber Extraction

Query processed in 0.8s · KP Forest Ordinance 2002, s.37 · Forest Act 1927, s.26

 Verified · 94%

STATUTES REFERENCED

 KP Forest Ordinance 2002 §37 Forest Act 1927 §26 PEPA 1997 §14 CrPC §154

ISSUE

Whether the unauthorized extraction, transportation, and sale of timber from a designated Reserved Forest in Mansehra District constitutes a criminal offence under the KP Forest Ordinance 2002 and the Forest Act 1927, and what penalties apply to the accused persons.

RULE

Under **Section 37, KP Forest Ordinance 2002**, any person who cuts, fells, girdles, lops, taps, or strips bark/leaves from trees in a Reserved Forest without authorization commits an offence punishable by imprisonment up to **3 years** and/or a fine of **Rs50,000–Rs500,000**. Section 26 of the **Forest Act 1927** additionally prohibits trespassing in reserved forests and offences thereunder carry penalties of Rs500 fine per tree. PEPA 1997 Section 14 further

 Key Statutes**§37 KP Forest Ordinance 2002**

Unlawful felling, girdling, lopping of trees in Reserved Forest. **Penalty: 3 years + Rs50K–Rs500K fine.**

§26 Forest Act 1927

Offences in Reserved Forests. Trespass, grazing, removal of forest produce. **Penalty: Rs500 per tree + confiscation.**

§14 PEPA 1997

Environmental Tribunal powers. Restoration orders, injunctions. **Penalty: up to Rs1M + restoration cost.**



INTELLIGENCE

Dashboard 3

AI Chat

LEGAL

Legal Analysis

Judiciary 47

OPERATIONS

Monitoring 23

Predict

Operations

PUBLIC

Citizen Reports 8

Permits

Legal Analysis Center

AI-Powered Legal Research · IRAC Framework · KP Forest Laws

KP Ordinance 2002

Forest Act 1927

PKT 15:57:23

KP Forest Ordinance 2002 §37

Forest Act 1927 §26

PEPA 1997 §14

CrPC §154

ISSUE

Whether the unauthorized extraction, transportation, and sale of timber from a designated Reserved Forest in Mansehra District constitutes a criminal offence under the KP Forest Ordinance 2002 and the Forest Act 1927, and what penalties apply to the accused persons.

RULE

Under **Section 37, KP Forest Ordinance 2002**, any person who cuts, fells, girdles, lops, taps, or strips bark/leaves from trees in a Reserved Forest without authorization commits an offence punishable by imprisonment up to **3 years** and/or a fine of **Rs50,000–Rs500,000**. Section 26 of the **Forest Act 1927** additionally prohibits trespassing in reserved forests and offences thereunder carry penalties of Rs500 fine per tree. PEPA 1997 Section 14 further empowers environmental tribunals to impose restoration orders.

APPLICATION

Based on satellite evidence from NASA FIRMS and GFW GLAD data showing 3.2 ha of cleared Reserved Forest area in Mansehra Block-7, combined with field patrol reports confirming timber loading operations, the facts constitute unlawful extraction. The accused were operating without a timber transit permit under Section 41 of the KP Forest Ordinance 2002. AI threat confidence at 92% corroborates physical evidence. Multiple accused persons qualify for joint prosecution under Section 34 PPC (common intention).

CONCLUSION

The extraction activity **constitutes a cognizable offence** under KP Forest Ordinance 2002 §37 read with Forest Act 1927 §26. Recommended action: (1) FIR registration under §37 KP Ordinance, (2) confiscation of timber and vehicles under §39, (3) application for Rs250,000 fine plus restoration order, (4) referral to Environmental Tribunal for PEPA 1997 proceedings. AI confidence: **94.2%**.

Unlawful felling, girdling, lopping of trees in Reserved Forest. **Penalty: 3 years + Rs50K–Rs500K fine.**

§26 Forest Act 1927

Offences in Reserved Forests. Trespass, grazing, removal of forest produce. **Penalty: Rs500 per tree + confiscation.**

§14 PEPA 1997

Environmental Tribunal powers. Restoration orders, injunctions. **Penalty: up to Rs1M + restoration cost.**

§22 Wildlife Act 1975

Protected species offences. Hunting, trapping, collection prohibited. **Penalty: 2 years + Rs200K fine.**

Legal Statistics

Conviction Rate (2024) 66.2%

Avg. Case Duration 4.8 months

Fines Collected (YTD) ₹3.76M

Outstanding Fines ₹9.52M

G

Home

DMs

More

Admin

GreenLawAI

Upgrade Plan

Find a conversation...

Slackbot

Huddles

Directories

Starred

Drag and drop important stuff here

Channels

all-greenlawai

new-channel

social

Slack works better when you use it together.

Invite teammates

Search GreenLawAI

all-greenlawai

MessagesAdd canvasFiles & linksNew+

GreenLawAI Alerts

GreenLawAI Autonomous Report

Analysis Report

Query: Show me the latest satellite evidence for forest crimes in Haripur

📖 Tactical Historical Intelligence

📡 **SATELLITE INTELLIGENCE: REAL-TIME DETECTION for Hazara**

A high-priority alert has been detected within the last 24 hours via tactical sensors.

📡 **CURRENT ALERT DETECTED (Today)**:

- **Date:** 2026-06-09
- **Location:** Abbottabad
- **Description:** Deforestation alert detected via GFW satellite. Confidence: HIGH
- **Source:** Global Forest Watch 🔗 [View Map](https://www.globalforestwatch.org/map/country/PAK/)

B I U | 🔗 | | | | </> | 📎

Message #all-greenlawai

+ Aa 😊 @ 📺 🎤 | 📄

Activate Windows
Go to Settings to activate Windows.



Compose



Inbox

987



Starred



Snooked



Sent



Drafts

15



Purchases

10



More

Labels



[CITIZEN REPORT] 🔥 Fire / Smoke — Mansehra Inbox x



alihinaali2022@gmail.com

to me ▾



CITIZEN FOREST REPORT

Report ID : CIT-20260608140854

Date/Time : 2026-06-08 14:08:54

Type : 🔥 Fire / Smoke

Location : Mansehra

Description:

fire detected

Citizen Contact: Anonymous

Submitted via GreenLaw AI — Hazara Forest Intelligence Hub